

MASTER WEB PAGE

User Guide & Reference Manual

Robotics & Automation Portfolio
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github.com/akshaykangude/Master-Web-Page

akshaykangude.github.io/Master-Web-Page

PRJ-001	Path Planning Algorithms Visualizer	BFS · DFS · Dijkstra · A* · RRT · RRT* · D* Lite · Potential Field
PRJ-002	LiDAR SLAM Simulator	360 raycasting · Occupancy grid · A* / Bug2 nav · Safety zones
PRJ-003	Robot Arm Simulator	N-DOF DH kinematics · IK · 5 motion profiles · Jacobian · Wrench calc

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1. Overview

Master Web Page is a GitHub Pages portfolio hosting three fully interactive, browser-based robotics and automation simulation demos — no installation required. Each demo is a faithful JavaScript port of the original Python desktop application, running the same algorithms and physics in the browser via WebAssembly (Pyodide) or pure JS.

Live URL	https://akshaykangude.github.io/Master-Web-Page
Repository	github.com/akshaykangude/Master-Web-Page
Stack	HTML · CSS · JavaScript · Pyodide (WebAssembly)
Hosting	GitHub Pages — free, auto-deploys on git push
Projects	3 interactive demos (path planning, SLAM, robot arm)

2. How to Use the Portfolio Page

Open the live URL in any modern browser (Chrome, Firefox, Edge, Safari). The home page shows all three projects as cards. Click **Run Live Demo** on any card to open that simulator.

Run Live Demo	Opens the interactive browser demo for that project.
Source	Opens the GitHub repository with the full Python source code.
PRJ-00X label	Project identifier. Green LED = demo is live and running.
Status bar	Bottom of each demo — shows current operation and errors.

NOTE: LiDAR SLAM loads a Python runtime (Pyodide) on first visit. This takes 5-15 seconds. Subsequent visits are faster due to browser caching.

3. Project 1 — Path Planning Algorithms Visualizer

An interactive grid where you can watch 8 different pathfinding algorithms explore the space between a start and goal point in real time. Draw walls, change grid size, and compare how each algorithm behaves.

Algorithms Available

BFS	Breadth-First Search — guarantees shortest path, explores uniformly.
DFS	Depth-First Search — fast but does not guarantee shortest path.
Dijkstra	Like BFS but weighted — always finds optimal path with costs.
A*	Dijkstra + heuristic — fastest to goal, standard in robotics.
RRT	Rapidly-exploring Random Tree — random sampling, good in open space.
RRT*	Asymptotically optimal RRT — rewires tree for shorter paths.
D* Lite	Dynamic A* — efficient replanning when obstacles change.
Potential Field	Attraction to goal + repulsion from walls — reactive navigation.

How to Use

1	Select an algorithm Click any algorithm button in the top toolbar.
2	Place start and goal Click Start then click a cell. Click Goal then click another cell.
3	Draw walls (optional) Click Wall then click-drag across cells to block paths.
4	Run Click Run . Watch the algorithm explore. Green = visited, yellow = path found.
5	Compare Click Clear to reset visited cells (keeps walls). Switch algorithm and run again to compare behaviour.

Controls

Run	Start the selected algorithm from current start/goal.
Clear	Clear visited cells and path — keep walls and start/goal.

Reset	Clear everything including walls.
Speed slider	Control animation speed from slow (educational) to instant.
Grid Size	Resize the grid — larger = more detail, slower on dense algos.

4. Project 2 — LiDAR SLAM Simulator

A full 2D LiDAR SLAM (Simultaneous Localisation and Mapping) simulator running Python physics via Pyodide. The robot emits 360 laser rays, builds an occupancy grid map from hits, navigates with A* or Bug2, and enforces proximity safety zones.

Toolbar Tools

Navigate	Manual control. Use arrow keys / WASD to drive the robot.
Go To	Click any point on the map — robot plans and follows a path automatically.
Bug2	Toggle Bug2 reactive navigation (see below). Works with Go To.
Circle	Add a circular obstacle. Set radius in the right panel.
Box	Add a rectangular obstacle. Set width/height in the right panel.
Wall	Draw a custom wall segment by clicking two points.
Move	Click and drag any obstacle to reposition it.
Resize	Click an obstacle then drag its handle to resize.
Delete	Click any obstacle to remove it.
Room Size	Drag the room boundary to resize the environment.
Record	Record LiDAR scans for scan-matching / localisation calibration.
Save Map	Download the current map + obstacle layout as a JSON file.
Load Map	Load a previously saved map JSON file.
Calibrate	Enter calibration mode — record reference scans at known poses.
Clear Map	Reset the occupancy grid (keep obstacles).

Right Panel — Settings

Proximity Safety Zones	Three concentric zones: Green (safe radius), Yellow (warn radius), Red (stop radius). Sliders set the radius — diameter shown next to it. Robot halts forward motion when forward LiDAR reads inside Red zone.
LiDAR Range	Maximum range of each laser ray in metres.
Rays / Scan	Number of rays per 360 scan (60-360). More = denser map.
FOV	Field of view in degrees. 360 = full circle.
Scan Hz	Scan frequency. Higher = map updates faster.
Robot Radius	Physical size of the robot for collision checking.
Move Speed	Speed in metres per step for autonomous navigation.

New Object Size

Default size for the next circle/box you place.

A* Navigation (default Go To mode)

When you click Go To, the planner runs A* on the occupancy grid with up to 3 clearance retries (full robot radius, half, then zero). If all three fail, the goal is marked UNREACHABLE in red.

Bug2 Navigation

Enable with the Bug2 toolbar button, then click Go To. Bug2 is a reactive algorithm — it does not use the grid at all:

M-Line	Straight line from robot start to goal (shown as purple dashed line).
MOVE_TO_GOAL	Robot drives directly toward goal along the M-line.
Obstacle hit	When LiDAR detects obstacle within safety distance, switches to BOUNDARY_FOLLOW. Hit point H is marked with an orange dot.
BOUNDARY_FOLLOW	Robot follows the obstacle boundary (8-direction scan, picks best clear angle). Counts steps — status panel shows boundary step count.
M-line re-cross	When robot crosses the M-line at a point closer to goal than H, it resumes MOVE_TO_GOAL mode.
No Path	If robot returns to H without finding a closer crossing, OR boundary steps exceed 2000, goal is marked NO PATH — obstacle completely surrounds the goal.

Keyboard Controls (Navigate mode)

W / Arrow Up	Move forward (blocked if forward LiDAR < Red zone threshold).
S / Arrow Down	Move backward (blocked if backward LiDAR < Red zone threshold).
A / Arrow Left	Rotate counter-clockwise.
D / Arrow Right	Rotate clockwise.
Q	Rotate counter-clockwise (alternative).
E	Rotate clockwise (alternative).
Enter	After scan matching — accept relocalisation and jump to matched pose.

5. Project 3 — Robot Arm Simulator

A real-time 3D N-DOF robot arm simulator using standard Denavit-Hartenberg kinematics, damped least-squares IK, and 5 motion profile types. Switch between a 3-DOF simple arm and a 6-DOF articulated arm. Five tabs give full control over the arm, its motion, graphs, edit mode, and live calculations.

3D Viewport Controls

Left-drag	Orbit camera around the arm.
Right-drag	Pan the camera up/down/left/right.
Scroll wheel	Zoom in / out.
R key	Reset arm to home pose.
Grid button	Toggle ground grid visibility.
Axes button	Toggle XYZ world axes.
Trail button	Toggle end-effector motion trail.
Target button	Toggle IK target marker (green = reached, red = not).
Reset View	Snap camera back to default angle.

Tab 1 — Control

The main tab for driving the arm and running trajectories.

Robot Config	Switch between RA-3 Simple Arm (3-DOF) and RA-6 Articulated Arm (6-DOF).
Joint Sliders	Drag any slider to move that joint in real time. Check the lock checkbox to exclude a joint from IK and trajectory motion.
IK Target X/Y/Z	Enter a target position in mm and click Solve IK. The arm moves to the nearest reachable pose. Convergence status and error shown below.
Trajectory Profile	Quintic (smooth polynomial), Linear (constant speed), or Motion (uses whatever profile is set in the Motion tab).
Timing Mode	Synced = auto-computes duration from motor speeds. Time-Opt = each joint at its own max speed. Manual = use the Duration slider.
Play	Animate from current pose to last IK solution using the selected profile.
Reset Home	Return all joints to their home angles.
Motor Load	Live bar chart showing each motor's speed as % of max rated RPM, visible while a trajectory plays.
Live Stats	EE position, reach range, max EE speed, total mass, IK error, active profile, locked joint count.

Tab 2 — Edit

Live-edit every parameter of every joint. Changes apply instantly to the 3D view.

Geometry	Link Length (mm) — also syncs the DH 'a' parameter. Link Diameter (mm) — visual only. Mass (kg) — used in torque calc.
DH Parameters	a = link length offset. d = link offset along Z. theta_offset = constant added to joint angle.
Rotation Plane (alpha)	Dropdown preset: XY plane (a=0), XZ plane (a=-90), YZ plane (a=+90), or custom numeric input. This is the DH alpha parameter — changes which plane the joint rotates in. Description updates to explain the effect. NOTE: alpha on joint N affects how joint N connects to N+1, NOT joint N-1.
Limits	Min/max angle limits in degrees. Arm clamps automatically.
Direction	Dir: + or Dir: - toggles rotation sign. Use to flip joint direction.
Motor	Output RPM, Rated Torque (Nm), Motor Weight (kg) — used in speed/torque calcs.
+ Add Joint	Appends a new default joint at the end of the chain.
X Remove	Removes that joint from the chain. Minimum 1 joint.

Tab 3 — Motion

Select and tune the motion profile used when Play is pressed.

7-Segment	Jerk-limited S-curve. 7 editable segments: J+ ramp, const accel, J- ramp, cruise, J- decel start, const decel, J+ decel end.
Trapezoidal	Classic 3-segment: acceleration, cruise, deceleration.
S-Curve 5	5-segment smooth curve with sinusoidal transitions.
Sinusoidal	Cycloidal profile — very smooth, zero jerk at ends.
Mod. Sine	Modified sine — lower peak acceleration than pure sinusoidal.
Segment sliders	Each segment has a % slider. Sum must equal 100% (shown in green). Sliders auto-normalise.
Preview canvas	Live Pos / Vel / Jerk waveform with colour-coded segment shading.
Apply button	Activates this profile for the next Play. Selecting any profile card or moving a slider auto-activates.

Tab 4 — Graph

Shows joint position, velocity, acceleration, or jerk plotted over time from the last trajectory played.

Joint toggles	Click J1-J6 buttons to show/hide individual joint lines. Colour matches the 3D viewport joint colour.
View modes	Pos (degrees), Vel (deg/s), Acc (deg/s ²), Jerk. Switch anytime after play.
Segment lines	Dashed vertical lines show motion profile segment boundaries.

Time axis	X axis in seconds. Y axis auto-scales to the data range.
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Tab 5 — Calc

Live numerical calculations updating with every joint movement.

Forward Kinematics	Current EE X, Y, Z in mm and total reach distance from base.
Inverse Kinematics	Last IK solve error in mm and convergence status.
Geometric Jacobian	Full 6xN matrix (rows: vx,vy,vz,wx,wy,wz; cols: joints). Built analytically from joint axes and positions.
Max EE Speed	Upper-bound EE speed in mm/s and m/s, with per-joint contribution breakdown.
Reach Limits	Min/max reachable distance from base via random joint sampling.
DH Parameters	Table of a, alpha, d, theta_offset for each joint, plus live current joint angle highlighted in amber.
DH Transform Matrices	Full 4x4 homogeneous transform $T(0 \rightarrow i)$ for each joint at the current pose. Rotation elements highlighted in blue.
EE Wrench ($J^T * F$)	6-component force/torque vector $[F_x, F_y, F_z, T_x, T_y, T_z]$ at EE from gravity and payload. Then $J^T * F$ gives per-joint torques. Payload mass adjustable via slider (0-20 kg).
Static Torque	Required torque per joint from link gravity + payload vs rated motor torque. Green = within limits, Red = overloaded.

6. GitHub Deployment Guide

The site is hosted on GitHub Pages — free static hosting that auto-deploys every time you push to the main branch. No build step required.

1	Create the GitHub repository Go to github.com → click + → New repository Repository name: Master-Web-Page Visibility: Public Leave everything else unchecked. Click Create repository.
2	Initialise and push from PowerShell Open PowerShell in your portfolio-site folder: <pre>cd C:\Users\AKSHAY\Documents\Git\portfolio-site\portfolio-site git init git add . git commit -m "Initial deploy" git remote add origin https://github.com/akshaykangude/Master-Web-Page.git git branch -M main git push -u origin main</pre>
3	Generate a Personal Access Token (PAT) GitHub no longer accepts your account password for git push. Go to: github.com -> Profile picture -> Settings -> Developer settings -> Personal access tokens -> Tokens (classic) -> Generate new token (classic) Tick the repo checkbox. Set expiration. Click Generate token. Copy the token immediately — paste it as the password when PowerShell asks.
4	Enable GitHub Pages Go to your repo on GitHub. Settings -> Pages (left sidebar). Source: Deploy from a branch Branch: main / Folder: / (root) -> Save
5	Visit your live site Wait ~2 minutes then open: https://akshaykangude.github.io/Master-Web-Page

7. Updating Your Site

Every git push to main triggers an automatic GitHub Pages rebuild. Changes go live in about 60 seconds.

Standard update workflow

```
cd C:\Users\AKSHAY\Documents\Git\portfolio-site\portfolio-site

# Make your changes to files, then:
git add .
git commit -m "describe what you changed"
git push
```

Adding a new project

1. Create folder	Add projects/your-project-name/ inside portfolio-site/
2. Add index.html	The demo page goes in that folder.
3. Link from home	In portfolio-site/index.html, add a new PRJ-00X panel card with a Run Live Demo button pointing to projects/your-project-name/index.html
4. Push	git add . && git commit -m "Add new project" && git push

Checking deploy status

Go to your repo on GitHub -> Actions tab. You will see a workflow run called 'pages build and deployment'. Green tick = live. Yellow = building. Red = error.

Common issues

Site shows 404	Wait 2-3 minutes after first push. Check that index.html is at the root of the repo, not in a subfolder.
Project links broken	The paths in index.html use relative links. Make sure your project folders are named exactly as referenced.
Auth fails on push	Use Personal Access Token as password, not your GitHub password.
Old version showing	Hard refresh: Ctrl+Shift+R in browser to bypass cache.
Pyodide demo stuck	LiDAR SLAM needs internet to load Pyodide CDN. Check browser console (F12) for errors.

8. Keyboard and Mouse Quick Reference

Path Planning Visualizer

Click cell	Place start, goal, or wall depending on selected tool.
Click+drag	Draw/erase multiple wall cells at once.
Run	Start algorithm animation.
Clear	Clear visited cells — keep walls and start/goal.
Reset	Clear everything.
Speed slider	Slow down animation for educational viewing.

LiDAR SLAM Simulator

W / Arrow Up	Move robot forward.
S / Arrow Down	Move robot backward.
A / Arrow Left	Rotate counter-clockwise.
D / Arrow Right	Rotate clockwise.
Q / E	Rotate left / right (alternative keys).
Go To + click	Set autonomous navigation goal (A* or Bug2).
Bug2 button	Toggle Bug2 reactive navigation mode.
Enter	Accept scan-match relocalisation result.
Record + Save/Load	Record scans, save map JSON, load map JSON.

Robot Arm Simulator

Left-drag viewport	Orbit camera.
Right-drag viewport	Pan camera.
Scroll	Zoom in / out.
R key	Reset arm to home pose.
Joint sliders	Move individual joints in real time.
Lock checkbox	Exclude joint from IK / trajectory.
Solve IK	Compute joint angles for target X/Y/Z.
Play	Animate trajectory to last IK solution.
Reset Home	Return all joints to home angles.
Edit tab -> alpha	Change rotation plane: XY (a=0), XZ (a=-90), YZ (a=+90).
Edit tab -> dir	Flip joint rotation direction (+/-).

+ Add Joint	Append new joint to chain.
X Remove	Delete joint from chain.
Motion tab	Select and tune 7-Seg / Trap / S-Curve / Sine / Mod-Sine profile.
Graph tab	View Pos/Vel/Acc/Jerk graphs after each trajectory play.
Calc tab	Live Jacobian, DH matrices, wrench, torque calculations.

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